时钟之迹

STM32F407 时钟配置与软件延时

STM32F407ZGT6
Keil MDK / STM32CubeMX

168 MHz



1 ms

HAL 时基周期
HAL_InitTick 配置

01 Set　02 Get　03 Config　04 Delay

Set

STD `Reset → SystemInit() → SetSysClock() → __main → main()`

HAL `Reset → SystemInit (基础配置) → __main → main()`

`HAL_Init() → SystemClock_Config()` `HAL_RCC_OscConfig() → HAL_RCC_ClockConfig()`

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fVCO = HSE ÷ M × N

HAL：8 ÷ 8 × 336 = 336 MHz

STD：25 ÷ 25 × 336 = 336 MHz

SYSCLK = 336 ÷ 2 = 168 MHz

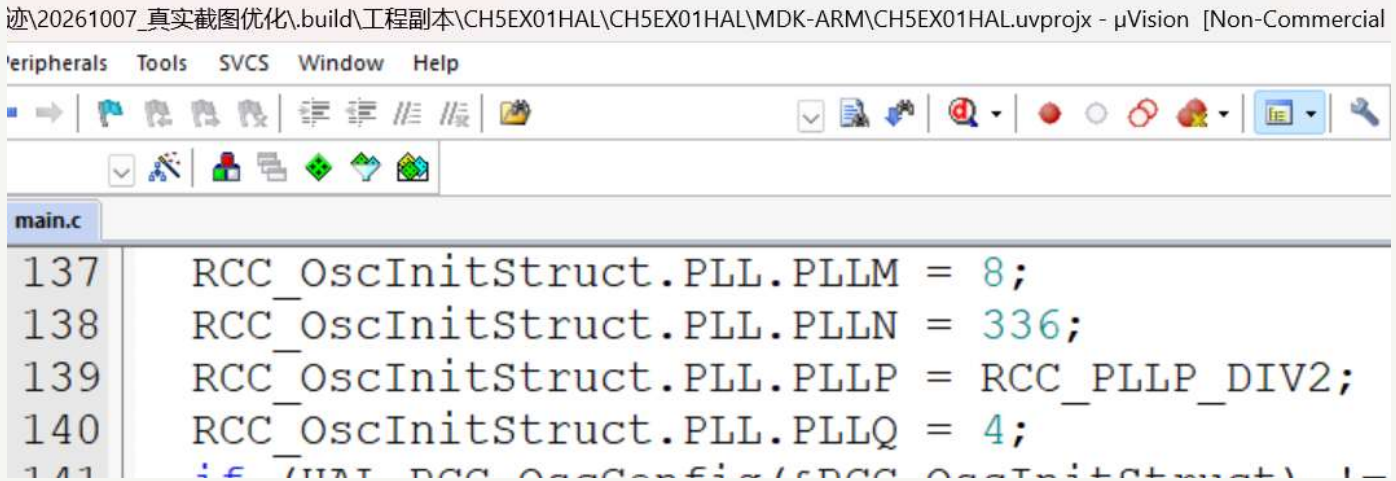
PLLQ 支路

原HAL Q=4：84 MHz > 75 MHz

建议Q=7：48 MHz（STD原值）

STD改接8 MHz晶振：

`HSE_VALUE=8000000, PLL_M=8`



AHB ÷1：HCLK = 168 MHz

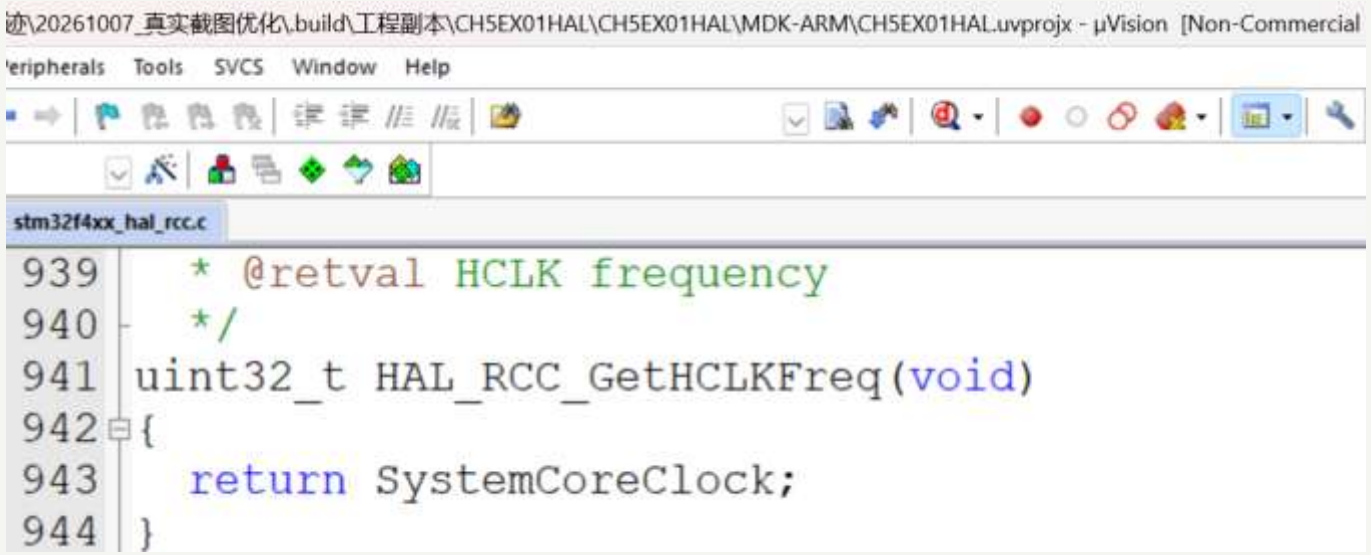
APB1 ÷4：42 MHz；APB2 ÷2：84 MHz

168 MHz、2.7–3.6 V：Flash 5 WS，Scale 1

Get

时钟	STD 结构成员	HAL 可用读取接口	MHz（推算）
SYSCLK	SYSCLK_Frequency	HAL_RCC_GetSysClockFreq()	168
HCLK	HCLK_Frequency	HAL_RCC_GetHCLKFreq()	168
PCLK1	PCLK1_Frequency	HAL_RCC_GetPCLK1Freq()	42
PCLK2	PCLK2_Frequency	HAL_RCC_GetPCLK2Freq()	84

```
RCC_GetClocksFreq(&RCC_Clocks); // Hz
```



频率信息

寄存器+HSE宏：计算频率

SystemCoreClock：HCLK软件记录

MCO+仪器：硬件实测频率

读源 → 算SYSCLK → 总线分频

改频后，同步HSE宏与软件记录

HAL_GetTick()：返回ms计数

Config

`T = (LOAD + 1) / fSysTick = ticks / HCLK`

24位递减计数；本例CLKSOURCE=1取HCLK，置0时可取HCLK/8。

HCLK = 168 MHz	STD • 10 ms	HAL • 默认1 ms
ticks	1 680 000	168 000
LOAD = ticks − 1	1 679 999	167 999

STD

```
SysTick_Config(
RCC_Clocks.HCLK_Frequency / 100);
```

HAL_InitTick()

```
HAL_SYSTICK_Config(
SystemCoreClock /
(1000U / uwTickFreq));
```

HAL_Init / 换频后重配，uwTickFreq=1

core_cm4.h

```
2027   }
2028
2029   SysTick->LOAD  = (uint32_t)(ticks - 1UL);
2030   NVIC_SetPriority (SysTick_IRQn, (1UL << __NVIC_
2031   SysTick->VAL   = 0UL;
2032   SysTick->CTRL  = SysTick_CTRL_CLKSOURCE_Msk |
2033                    SysTick_CTRL_TICKINT_Msk   |
2034                    SysTick_CTRL_ENABLE_Msk;
2035   return (0UL);
```

校验24位范围 → LOAD=ticks−1 → VAL=0 → 使能时钟/中断 → 周期重装

Delay

STD • Delay(10) ≈100 ms

```
static __IO uint32_t uwTimingDelay;
uwTimingDelay = n;
while (uwTimingDelay != 0);
// __IO = volatile
```

10 ms中断：TimingDelay_Decrement非零减1

Usermain.c

```
83 void Delay(__IO uint32_t nTime)
84 {
85   uwTimingDelay = nTime;
86
87   while(uwTimingDelay != 0);
88 }
```



HAL • HAL_Delay(1000) ≈1 s

```
start=HAL_GetTick(); wait=Delay;
if (wait<HAL_MAX_DELAY)
wait += uwTickFreq;
while (HAL_GetTick()-start<wait);
```

1 ms中断：HAL_IncTick累加uwTick

stm32f4xx_hal.c

```
312 __weak void HAL_IncTick(void)
313 {
314   uwTick += uwTickFreq;
315 }
```



主循环阻塞；计数由中断推进，屏蔽中断会停等。两者均非精密定时。

各时钟原理与应用

时钟	原理与应用
HSI 16 MHz	内部RC；复位启动、故障后备
HSE	外部晶振/时钟；PLL基准
PLL	先分频再倍频；合成高速时钟
SYSCLK	选HSI/HSE/PLL；系统主时钟
HCLK / PCLK	总线分频；CPU、AHB/APB外设
LSE 32.768 kHz	外部低速晶振；RTC日历
LSI ≈32 kHz	内部低速RC；IWDG、RTC备用
SysTick	24位递减；软件节拍、阻塞延时
TIM / PWM	APB÷1时TIM=PCLK；其余2×PCLK 硬件定时/PWM；本例84/168 MHz
外设门控	RCC使能位；外设访问及节能

```
__HAL_RCC_GPIOE_CLK_ENABLE();
```

STD • Delay(10)

每轮约100 ms

每5轮翻转：间隔约500 ms

完整亮灭周期：约1 s

HAL • HAL_Delay(1000)

每轮、翻转间隔约1 s

完整亮灭周期：约2 s

Index 回卷

`uint16_t; Index%5==0` 时翻转

65535 → 0

两者均满足%5==0

连续两轮翻转，间隔约100 ms